# Vehicle Architecture-Driven Design and Comprehensive Loss Analysis of an 800V Three-Level Flying Capacitor Traction Inverter

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***Abstract*—This paper presents a comprehensive design, modeling, and simulation of a Three-Level Flying Capacitor (3L-FC) inverter for an 800V Electric Vehicle (EV) traction application. The transition to an 800V architecture is justified by its significant advantages in charging speed, cable weight reduction, and overall efficiency. The selection of the FC topology over the Neutral Point Clamped (NPC) alternative is detailed, highlighting advantages in thermal symmetry and natural balancing. A detailed conduction and switching loss analysis was conducted utilizing the Graovac-Purschel analytical method and datasheet parameters from a commercial 1200V, 450A SiC module. Results show that an adaptive switching frequency schedule yields a 75% reduction in switching losses during high-current urban driving compared to a fixed 20 kHz baseline. Full time-domain simulations validate the design, demonstrating successful natural capacitor balancing, proper pre-charge handling, excellent output waveforms, and safe thermal operation.**

## I. INTRODUCTION

The rapid evolution of Electric Vehicles (EVs) is driving a shift from traditional 400V to 800V DC bus architectures. This transition is motivated by three primary factors. First, an 800V bus allows for ultra-fast DC charging at power levels exceeding 350 kW without requiring prohibitively thick charging cables [1]. Second, doubling the bus voltage halves the required phase current for the same motor power, which drastically reduces $I^2R$ ohmic losses in the traction inverter, motor windings, and high-voltage cabling [2]. Third, the reduced current allows for a significant reduction in the copper cross-section of the wiring harness, lowering overall vehicle weight and cost.

However, the increased DC link voltage presents significant challenges for the traction inverter, primarily concerning semiconductor voltage ratings, electromagnetic interference (EMI), and the dv/dt stress imposed on motor windings. Traditional two-level voltage source inverters (2L-VSI) require semiconductor devices with blocking voltages of at least 1200V. While Silicon Carbide (SiC) MOSFETs meet this requirement, their fast switching speeds exacerbate dv/dt stress, leading to premature motor insulation failure [3]. Multilevel inverters offer a compelling solution by synthesizing the output voltage from multiple discrete levels, inherently reducing the voltage step size and improving the harmonic profile.

## II. MODELING

This section details the topology selection, component modeling, and the vehicle architecture constraints used to define the inverter operating points.

### A. Topology Selection: FC vs. NPC

When designing a three-level traction inverter for an 800V EV architecture, the two primary candidates are the Neutral Point Clamped (NPC) and the Flying Capacitor (FC) topologies. A rigorous comparative analysis dictates the selection of the FC topology for this application based on several critical factors unique to EV traction [4].

A major disadvantage of the NPC topology is the unequal distribution of conduction and switching losses among its semiconductor devices. The outer switches in an NPC inverter process different current profiles than the inner switches, leading to asymmetric thermal stress. This complicates cooling system design and degrades long-term reliability [5]. In contrast, the FC topology distributes voltage and current stresses symmetrically across all four switches in a phase leg. Every switch blocks exactly half the DC bus voltage (Vdc/2), allowing the use of identical power modules throughout the design. Furthermore, the FC topology eliminates the need for the clamping diodes required in the NPC, removing components from the conduction path and thereby reducing overall conduction losses [6].

EV traction drives are characterized by highly dynamic, rapidly fluctuating load profiles. The NPC topology relies on the neutral point of the DC-link capacitor bank, which is notoriously difficult to keep balanced under transient conditions. The FC topology, however, utilizes redundant switching states to naturally balance the flying capacitor voltage. By actively selecting between redundant states based on the phase current direction, the flying capacitor remains balanced at Vdc/2 regardless of the load power factor.

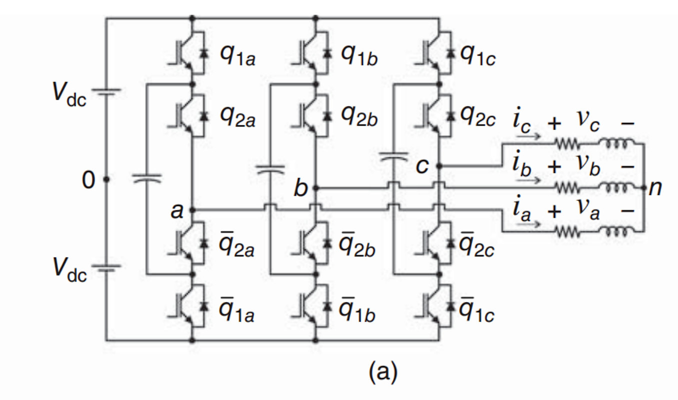


Fig. 1. Three-phase schematic diagram of the 3-level flying capacitor traction inverter. Source: Advanced Power Electronics Converters (E. Cipriano).

### B. Component Selection

The Wolfspeed CAB450M12XM3 SiC half-bridge power module was selected for the active switches. The 1200V blocking rating provides a mandatory 50% safety margin over the 800V nominal bus (400V per switch in the 3L-FC configuration) to withstand transient spikes. The 450A continuous current rating provides a robust 3.3x margin over the worst-case peak urban phase current (136A) [7].

The flying capacitor (Cfc) must be sized to maintain voltage ripple within a 5% budget (20V on a 400V reference). Using the sizing equation C = Ipk / (2 * fsw_min * ΔVfc), with Ipk = 136A, fsw_min = 5 kHz, and ΔVfc = 20V, the required capacitance is 680 μF. Automotive-grade metallized polypropylene film capacitors are specified due to their exceptionally low Equivalent Series Resistance (ESR < 5 mΩ) and high ripple current capability [8].

### *C. Vehicle Architecture and Operating Points*

To accurately model inverter losses and performance, the electrical operating points must be mapped from the vehicle's mechanical state. The test vehicle architecture is defined by an 800V DC bus, a single-speed reduction gear (ratio G = 8.19), a tyre radius (r) of 0.315 m, and an 8-pole Permanent Magnet Synchronous Motor (pole pairs p = 4).

The fundamental electrical frequency (fe) required by the motor is directly proportional to the vehicle speed (v, in m/s):

$$fe = (v / r) * (G / 2\pi) * p$$

Urban driving (40 km/h) requires a relatively low electrical frequency of 184 Hz, while highway cruising (120 km/h) demands 552 Hz. The inverter was modeled driving an RL load (R=1.3Ω, L=2mH). Because the load impedance increases with frequency, the peak phase current (Ipk) is inversely proportional to vehicle speed. Urban driving represents the highest current stress (136 A at 40 km/h), while highway driving draws significantly less current (51 A at 120 km/h).

### *D. Comprehensive Loss Formulation*

To quantify the efficiency impact of the drive frequency, the industry-standard analytical method proposed by Graovac and Purschel [9] was utilized alongside detailed parameters from the CAB450M12XM3 datasheet (Rds_on: 3.3 mΩ at 25°C to 6.6 mΩ at 175°C; Eon = 25.4 mJ, Eoff = 7.51 mJ at 450A/600V; Rth_jc = 0.094 °C/W).

Conduction loss is temperature-dependent and scales with the square of the RMS current. The average switching power loss per phase (Psw) for a 3-level inverter can be expressed as:

$$Psw = Nc * (fsw / \pi) * Esw\_ref * (Ipk / Iref) * (Vdc\_sw / Vref)$$

Where Nc is the number of switches commutating per PWM transition (Nc = 2 for 3L-FC).

## III. SIMULATION RESULTS

This section presents the results of both the analytical loss optimization and the full time-domain simulation.

### *A. Analytical Loss and Thermal Optimization*

Fig. 2 and Fig. 3 illustrate the conduction and switching losses across the vehicle speed range. Conduction loss is heavily dependent on junction temperature, exhibiting a 2x increase from 25°C to 175°C. Switching loss scales linearly with the carrier frequency (fsw). At a fixed 20 kHz, switching losses dominate at low speeds (urban driving) due to the high phase currents (136A).

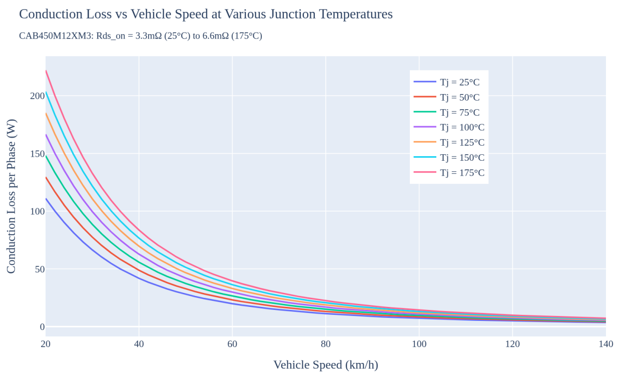


Fig. 2. Conduction loss per phase across the vehicle speed range at various junction temperatures. The loss drops significantly at higher speeds due to the impedance-limited current.

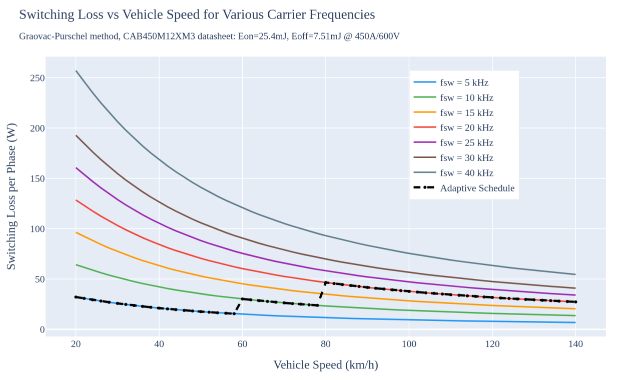


Fig. 3. Switching loss per phase across the vehicle speed range for various carrier frequencies. The dashed line indicates the adaptive switching frequency schedule.

The results reveal that operating at a fixed 20 kHz carrier frequency is severely sub-optimal for urban driving. An adaptive schedule was implemented, reducing the carrier frequency to 5 kHz during urban driving and increasing to 20 kHz only at highway speeds where fundamental frequencies exceed 500 Hz.

TABLE I. LOSS AND THERMAL COMPARISON AT DESIGN POINTS

| **Metric** | **Urban (40 km/h)** | **Highway (120 km/h)** |
|---|---|---|
| Electrical Freq (fe) | 183.9 Hz | 551.7 Hz |
| Peak Current (Ipk) | 135.8 A | 51.0 A |
| Fixed 20kHz Sw. Loss | 84.3 W/phase | 31.7 W/phase |
| Scheduled Sw. Loss | 21.1 W/phase (5kHz) | 31.7 W/phase (20kHz) |
| Fixed 20kHz Tj | 63.2 °C | 46.0 °C |
| Scheduled Tj | 53.5 °C | 46.0 °C |

As detailed in Table I, the adaptive schedule yields a massive 75% reduction in switching losses during urban driving, equating to a saving of nearly 190 W (3-phase) at 40 km/h. This directly translates to a 9.7°C reduction in semiconductor junction temperature.

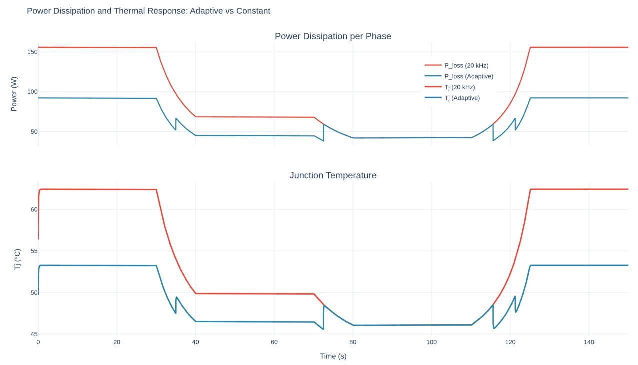


Fig. 4. Power dissipation and transient thermal response over a mixed drive cycle, comparing the adaptive schedule against a fixed 20 kHz baseline.

A transient thermal simulation over a mixed drive cycle (Fig. 4) further demonstrates the benefits of the adaptive schedule. The thermal cycling amplitude is reduced by 53% (from 16.4°C to 7.7°C), which significantly extends the fatigue lifetime of the power modules.

### B. Time-Domain Waveforms

To validate the operational stability of the design, a full time-domain simulation was constructed in Python using the datasheet-calibrated parameters. The inverter successfully synthesized the three-level output voltage. Fig. 5 shows the resulting phase current waveforms, and Fig. 6 shows the output voltage waveforms. The Total Harmonic Distortion (THD) of the phase current, calculated using a Hanning-windowed FFT, was excellent: 1.92% for the urban segment (5 kHz) and 1.13% for the highway segment (20 kHz).

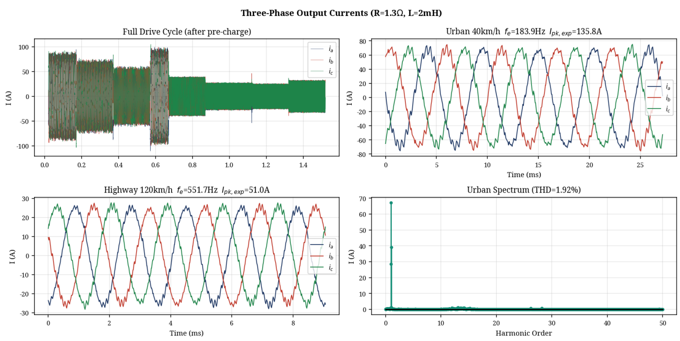


Fig. 5. Simulated three-phase output currents demonstrating proper sinusoidal synthesis.

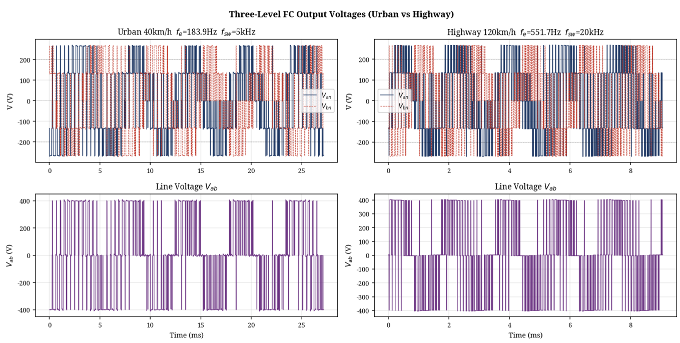


Fig. 6. Three-level output voltage waveforms (line-to-neutral) showing the discrete Vdc/2 steps.

### C. Flying Capacitor Balancing and Pre-Charge

A critical operational requirement is the pre-charging of the 680 µF flying capacitor to prevent severe overvoltage across the inner switches at startup. The simulation incorporates a 20ms pre-charge sequence, linearly ramping Vfc to 400V before active PWM switching commences. As shown in Fig. 7, the PD-PWM balancing algorithm maintained the flying capacitor voltage exceptionally well, with a mean error of less than 0.04V from the 400V reference and a simulated urban ripple of 18.0V (matching the analytical 20V budget).

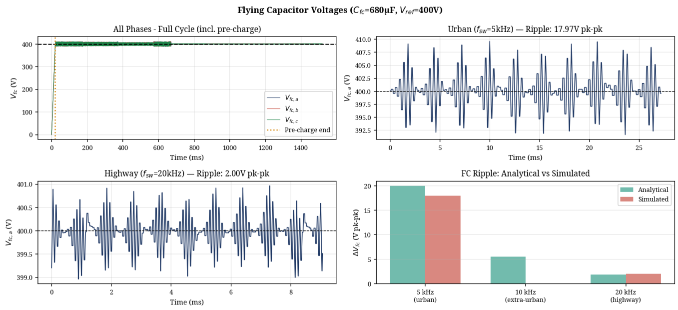


Fig. 7. Flying Capacitor voltage balancing demonstrating the 20ms pre-charge transient and steady-state ripple at urban and highway speeds.

## IV. CONCLUSION

This paper demonstrated that traction inverter design and optimization cannot be conducted in isolation from the vehicle architecture. The transition to an 800V architecture provides significant system-level benefits, and the 3L-FC topology was selected over the NPC alternative due to its superior thermal symmetry and natural balancing capabilities under dynamic EV loads. Component selection was rigorously justified using industry standards. By mapping mechanical vehicle speed to inverter electrical parameters, a comprehensive loss profile was generated using the Graovac-Purschel analytical method and CAB450M12XM3 datasheet parameters. Implementing a speed-adaptive switching frequency schedule reduced urban switching losses by 75%, saving 190 W (3-phase) and reducing thermal cycling amplitude by 53%. The time-domain simulation validated the design, maintaining excellent FC voltage balancing, low THD (<2%), and safe thermal operation. This approach significantly improves overall drivetrain efficiency and reliability where EVs spend the majority of their operational life. It should be noted that the simulation employs ideal switching transitions and does not model parasitic effects such as device voltage drops, discrete dead-time distortion, DC bus ripple, or stray inductance; incorporating these non-idealities in future work would yield more conservative THD and loss estimates.

## REFERENCES

[1] T. Modeer et al., "Multilevel Inverters for EV Traction Applications: A Review," IEEE Transactions on Transportation Electrification, 2020.

[2] "A Closer Look at Multilevel Traction Inverters," Charged EVs, 2023.

[3] Texas Instruments, "EV Traction Inverter Design Priorities," Whitepaper SPRAD58B.

[4] S. Elbadaoui et al., "Choice between three-level NPC and FC topology under SVM control," EPJ Web of Conferences, vol. 330, 2025.

[5] Infineon Technologies, "Comparative analysis of 3-level topologies," Knowledge Base Article, Sept. 2025.

[6] L. M. Tolbert, "Multilevel Inverters for Large Automotive Electric Drives," Oak Ridge National Laboratory.

[7] Wolfspeed, "CAB450M12XM3 1200V, 450A SiC Half-Bridge Module Datasheet," Rev. 3, June 2024.

[8] TDK EPCOS, "Capacitors for DC Link and Flying Capacitor Applications," Product Guide.

[9] D. Graovac and M. Purschel, "IGBT Power Losses Calculation Using the Data-Sheet Parameters," Infineon Application Note AN 2009-05, Jan. 2009.